

CYBER SMART

Understanding Cyber Security, Ethical Hacking & Cyber Forensics



Steve Sajan Jacob

Lead Software Engineer

BerriBot Inc

San Francisco, USA



About the Speaker



Steve Sajjan Jacob
Lead Software Engineer (Gen & Agentic AI) | Full Stack Developer

Started coding at 13. Now building AI solutions that transform businesses globally—from government systems to enterprise automation across US, UK, UAE, Singapore, Qatar, Kuwait, Oman, Tanzania, and India.

-  FULL STACK EXPERT
-  AI SPECIALIST
-  PATENT CO-AUTHOR

1

Global Recognition

Honored by Indian Parliament leaders including Hon. MP Dr. Shashi Tharoor, Hon. MoS Mr. Suresh Gopi, Hon. Ex-MoS Mr. Rajeev Chandrasekhar, and Hon. MP Mr. Hibi Eden.

2

International Engagement

Only Keralite student selected from India for Loyola University Chicago's Building Bridges initiative—engaged with Pope Francis on global issues.

3

Patent Co-Author

Co-author of the PATENT application - “METHODS FOR PROVIDING AN INTELLIGENT INTERVIEWING PLATFORM AND DEVICES THEREOF” being the first of its kind innovation in recruitment tech space

Professional Experience & Credentials



Global Projects

Delivered solutions for international clients across multiple continents, working with distributed teams and managing complex technical requirements at scale.



Domain Expertise

Web Development, Artificial Intelligence, Process Automation, Financial Technology, and E-Commerce platforms—building products that serve millions of users.



Industry Certifications

Certified by Google, IBM, and AWS in cloud computing, machine learning, and advanced web technologies—validating expertise with industry-recognised credentials.

 "Started coding early, built real products, works at scale"



Why Cyber Security Matters

Digital Life = Digital Risks

Every online interaction creates a potential vulnerability. From banking to social media, our entire lives are now digital targets.

Rising Cyber Crime

Global cyber crime damages are projected to reach \$10.5 trillion annually by 2025, making it more profitable than the global drug trade.

Data Breaches & Loss

Major corporations lose millions in breaches. Personal data theft can destroy lives. The average cost of a data breach is now over \$4 million.

Developer Responsibility

As future developers, you're not just building features—you're protecting users. Security must be built in, not bolted on.

What is Cyber Security?

Definition

Cyber security is the practice of protecting systems, networks, and programmes from digital attacks. These attacks typically aim to access, change, or destroy sensitive information, extort money from users, or interrupt normal business processes.

Understanding the Landscape

- **Threats:** Potential dangers that could exploit vulnerabilities (e.g., hackers, malware)
- **Vulnerabilities:** Weaknesses in a system that can be exploited (e.g., outdated software, weak passwords)
- **Attacks:** Actual attempts to breach security and cause harm (e.g., phishing campaign, DDoS)

📌 Real-World Analogy

Think of cyber security like protecting your home:

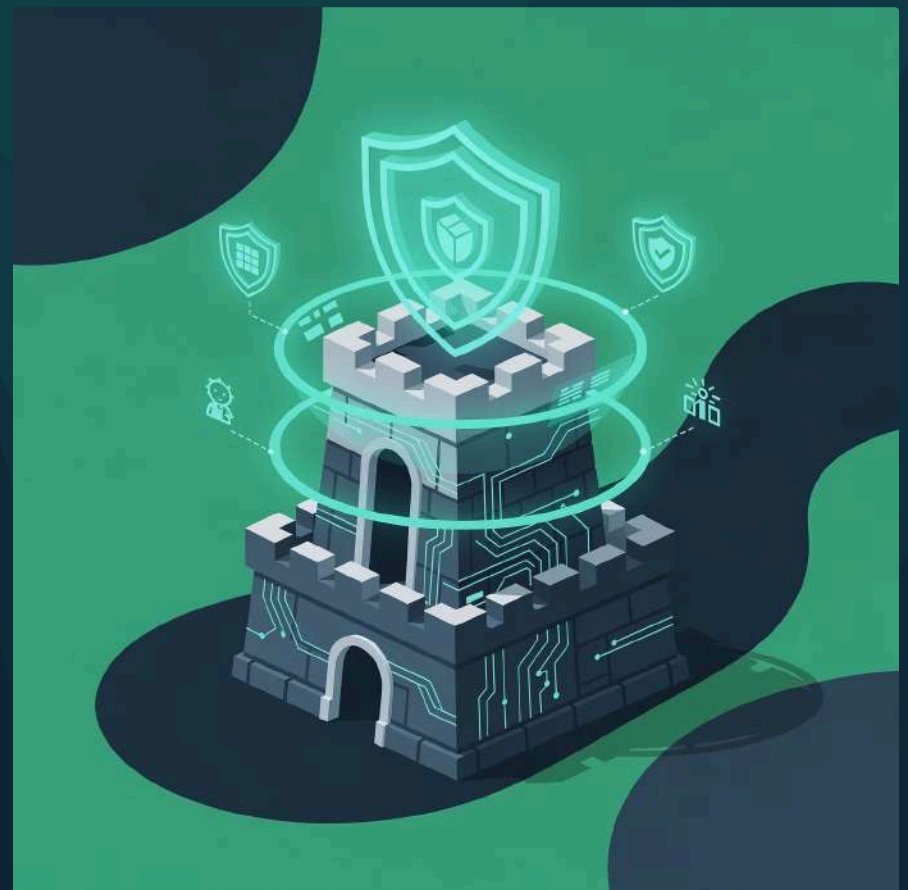
🏠 **Your house** = Your computer system

🔒 **Locks and alarms** = Firewalls and antivirus

🚪 **Weak door** = Vulnerability

👤 **Burglar** = Threat

💥 **Break-in** = Attack



Types of Cyber Threats

Understanding the enemy is the first step in defence. Today's cyber threats are sophisticated, constantly evolving, and can strike from anywhere in the world.



Malware

Malicious software designed to damage or gain unauthorised access to systems. Includes viruses, trojans, worms, and spyware that can steal data, corrupt files, or take control of devices.



Phishing

Fraudulent attempts to obtain sensitive information by disguising as trustworthy entities. Often via email, these attacks trick users into revealing passwords, credit card details, or personal information.



Ransomware

Malware that encrypts your files and demands payment for decryption. Recent attacks have crippled hospitals, schools, and major corporations, with ransoms reaching millions of pounds.



Social Engineering

Psychological manipulation to trick people into breaking security procedures. Attackers exploit human trust rather than technical vulnerabilities, making this one of the most effective attack vectors.



Insider Threats

Security risks from within an organisation, whether malicious or accidental. Employees with legitimate access can cause devastating damage, accounting for nearly 60% of all security incidents.



Core Security Principles: The CIA Triad

Every security strategy is built on three fundamental pillars. Understanding these principles helps you design secure systems from the ground up.

Confidentiality

Keeping secrets safe. Ensuring that sensitive information is accessible only to those authorised to view it.

Student Example: Your exam results should only be visible to you, your professors, and authorised staff—not your classmates.

Integrity

Maintaining accuracy. Ensuring that data remains accurate, complete, and unaltered during storage or transmission.

Student Example: Your marks in the university database should not be altered by anyone except authorised personnel through proper procedures.

Availability

Access when needed. Ensuring that authorised users have reliable access to information and systems when required.

Student Example: The university portal should be accessible 24/7 so you can register for courses, check results, or access resources whenever needed.

Networking Basics for Security

What is a Network?

A network is a collection of computers and devices connected to share resources and communicate. The internet is the world's largest network, connecting billions of devices globally.

Key Networking Concepts

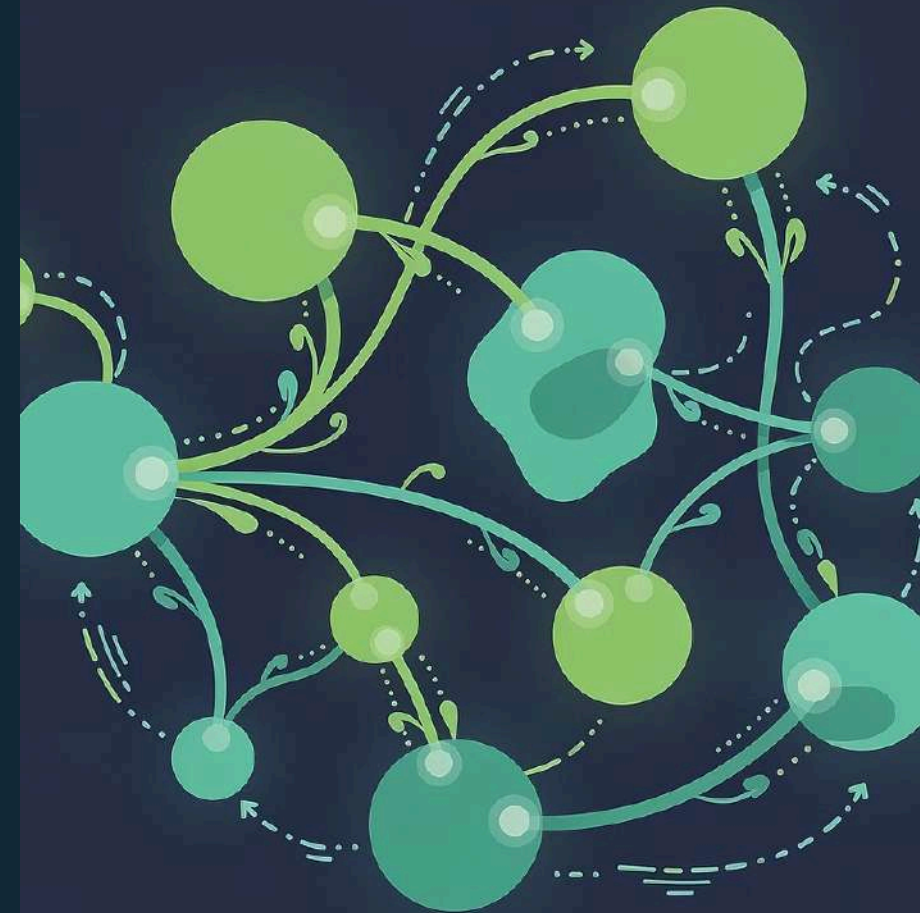
- **IP Address:** A unique identifier for each device on a network (like a postal address for your computer)
- **DNS (Domain Name System):** Translates human-readable names (google.com) into IP addresses computers understand
- **HTTP vs HTTPS:** HTTP is unencrypted web communication; HTTPS adds encryption for secure data transfer

Why It Matters in Security

Every security professional must understand networks because:

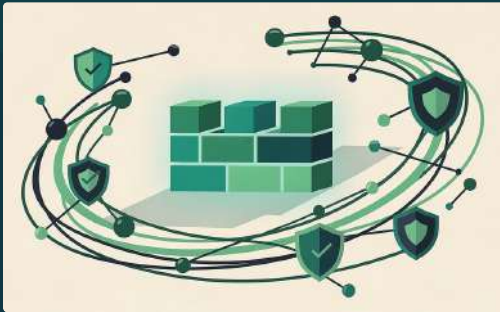
1. Attackers exploit network vulnerabilities to gain access
2. Data travels across networks and can be intercepted
3. Network monitoring helps detect suspicious activity
4. Proper network configuration is your first line of defence

 **Remember:** Understanding how data flows through networks helps you protect it at every step of its journey.



Network Protection Mechanisms

Multiple layers of defence work together to create a secure network environment. Each mechanism serves a specific purpose in protecting your digital assets.



Firewalls

Act as a barrier between trusted internal networks and untrusted external networks. They monitor and control incoming and outgoing traffic based on predetermined security rules, blocking potentially harmful connections.



Antivirus Software

Continuously scans systems for known malware signatures and suspicious behaviour. Modern antivirus solutions use machine learning to detect even previously unknown threats, providing real-time protection.



Access Control

Determines who can access which resources and what actions they can perform. Implements the principle of least privilege—users only get the minimum access necessary to perform their duties.



Authentication vs Authorisation

Authentication verifies who you are (username/password).

Authorisation determines what you can do (permissions). Both are essential—you need to prove your identity AND have appropriate permissions.

Web Applications & Security Risks

Web applications are the most common target for cyber attacks because they're publicly accessible and often handle sensitive data. Understanding these vulnerabilities is crucial for any developer.

Why Websites Are Prime Targets

Web applications are exposed to the internet 24/7, handle valuable user data, and often have multiple entry points. A single vulnerability can expose thousands or millions of users to risk.



SQL Injection

Attackers insert malicious SQL code into input fields to manipulate databases. This can expose entire databases, delete data, or grant unauthorised access. It remains one of the top web vulnerabilities.



Cross-Site Scripting (XSS)

Malicious scripts are injected into trusted websites, executing in victims' browsers. These scripts can steal session cookies, redirect users to malicious sites, or deface web pages.



Broken Authentication

Flaws in authentication systems allow attackers to compromise passwords, keys, or session tokens. Weak password policies, exposed session IDs, and poor credential management create these vulnerabilities.



Insecure APIs

Application Programming Interfaces (APIs) are gateways between systems. Poor API security can expose sensitive data, allow unauthorised actions, or provide entry points for larger attacks on infrastructure.



Safe Online Behaviour

Technology alone cannot protect you—your behaviour is the strongest defence. These practices should become second nature in your digital life.

O1

Strong Passwords

Use 12+ characters with uppercase, lowercase, numbers, and symbols. Never reuse passwords across sites. Consider using passphrases—longer phrases are easier to remember and harder to crack.

O2

Two-Factor Authentication (2FA)

Add a second verification step beyond passwords. Even if someone steals your password, they cannot access your account without the second factor (SMS code, authenticator app, or hardware key).

O3

Phishing Awareness

Verify sender addresses carefully. Hover over links before clicking. Be suspicious of urgent requests or unexpected attachments. When in doubt, contact the organisation directly through official channels.

O4

Email & Browsing Safety

Only visit HTTPS websites when entering sensitive data. Keep software updated. Use reputable browsers with security features. Be cautious about what you download and from where.



Quick Security Checklist

- ✓ Use a password manager
- ✓ Enable 2FA everywhere possible
- ✓ Keep all software updated
- ✓ Think before you click
- ✓ Use different passwords for different sites
- ✓ Back up important data regularly

What is Ethical Hacking?

Ethical Hacker vs Criminal Hacker

Both use similar techniques and tools, but the crucial difference lies in **permission, intent, and legality**.

Ethical hackers (white hat hackers) work with authorisation to identify and fix vulnerabilities. They protect systems and help organisations improve security.

Criminal hackers (black hat hackers) break into systems without permission for personal gain, causing damage, or stealing data. Their actions are illegal and harmful.

Why Companies Hire Ethical Hackers

- Find vulnerabilities before criminals do
- Test security systems under real-world conditions
- Comply with security standards and regulations
- Protect customer data and company reputation
- Save millions in potential breach costs

❏ **Critical:** Ethical hacking requires written permission, a defined scope, and full legal authorisation. Without these, you're breaking the law—no matter your intentions.



Phases of Ethical Hacking

Ethical hackers follow a structured methodology to assess security systematically. Each phase builds on the previous one, mimicking how real attackers operate.



Reconnaissance & Scanning

These initial phases are critical—attackers spend 80% of their time gathering information. The more they know about your systems, the easier it is to find weaknesses.

Footprinting & Information Gathering

Collecting publicly available information about the target:

- Company websites and social media
- Employee information and organisational structure
- Domain registration details (WHOIS)
- Public records and financial reports
- Technology stack and infrastructure details
- Physical location and network range

Attackers use this information to map the organisation's digital presence and identify potential entry points.

Enumeration & Active Scanning

Actively probing systems to discover detailed information:

- Open ports and running services
- Operating system versions
- Network topology and device types
- User accounts and access permissions
- Shared resources and weak configurations



Why Attackers Start Here

Information is power. The more an attacker knows, the more targeted and effective their attack can be. Good reconnaissance can reveal vulnerabilities without ever touching the target system.

System & Web Exploitation

Once vulnerabilities are identified, the exploitation phase demonstrates their real-world impact. Understanding these techniques helps you build more secure systems.



Malware Basics

Malicious software comes in many forms—viruses that replicate, trojans that disguise themselves, worms that spread automatically, and spyware that monitors activity. Modern malware is sophisticated, often using encryption to evade detection and multiple attack vectors simultaneously.



Privilege Escalation

After gaining initial access with limited permissions, attackers attempt to escalate privileges to gain administrator or root access. This allows them to install persistent backdoors, access sensitive data, and control the entire system. Exploiting misconfigured permissions is a common technique.



Web Exploitation

Web applications are complex systems with many potential vulnerabilities. Beyond SQL injection and XSS, attackers exploit file upload vulnerabilities, insecure direct object references, and authentication bypass techniques. Each vulnerability can lead to full system compromise.



Importance of Patching

Most successful attacks exploit known vulnerabilities that already have patches available. Regular updates and patch management are critical. The WannaCry ransomware attack affected 200,000+ computers precisely because they hadn't installed a patch released months earlier.

Wireless Network Security

Wi-Fi Encryption Standards

Wireless networks broadcast data through the air, making encryption essential:

- **WEP (Wired Equivalent Privacy):** Outdated and easily cracked—never use
- **WPA (Wi-Fi Protected Access):** Better than WEP but still vulnerable
- **WPA2:** Current standard, strong encryption when properly configured
- **WPA3:** Latest standard with enhanced security features and protection against brute-force attacks

Public Wi-Fi Risks

Public networks are convenient but dangerous:

1. Traffic can be intercepted easily
2. Fake "evil twin" networks can steal credentials
3. Man-in-the-middle attacks are simple to execute
4. Other users on the network may be malicious

Protection: Use VPNs on public networks and avoid accessing sensitive accounts.



Basic Wireless Attack Awareness

Packet Sniffing: Capturing data transmitted over wireless networks to steal information or analyse traffic patterns.

Rogue Access Points: Unauthorised Wi-Fi access points that provide network access or capture credentials from unsuspecting users.

Deauthentication Attacks: Forcing devices to disconnect from networks, potentially directing them to malicious networks.

 **Student Tip:** At home, use WPA2/WPA3, change default router passwords, hide SSID if possible, and regularly update router firmware.

Introduction to Cyber Forensics

Cyber forensics is the science of investigating digital crimes and incidents. When prevention fails, forensics helps identify attackers, understand what happened, and provide evidence for legal proceedings.

What is Cyber Forensics?



Cyber forensics (or digital forensics) is the process of collecting, preserving, analysing, and presenting digital evidence in a manner that's legally admissible. It involves recovering data from computers, networks, mobile devices, and other digital storage to reconstruct events and identify perpetrators.

When is it Used?



Forensic investigations are triggered by various incidents: data breaches requiring root cause analysis, criminal investigations involving digital evidence, intellectual property theft, employee misconduct cases, fraud investigations, and compliance violations. Any incident where digital evidence might exist requires forensic expertise.

Role in Cyber Crime Investigation



Forensic experts reconstruct timelines of events, identify attack vectors, trace attacker movements through systems, recover deleted or encrypted data, and provide expert testimony in court. Their work often determines whether cases can be prosecuted successfully and helps organisations prevent future incidents.

Digital & Mobile Forensics Basics

Modern forensics encompasses various specialised areas, each requiring specific expertise and tools. As our digital footprint expands, so does the scope of forensic investigation.



File System Analysis

Examining how data is stored, organised, and deleted on storage devices. Forensic analysts can recover deleted files, identify file modifications, analyse file metadata, and detect attempts to hide or encrypt data. Different file systems (NTFS, FAT, ext4) require different analysis techniques.



Log Analysis

System logs record user activities, network connections, application events, and security incidents. Analysts examine logs to establish timelines, identify anomalies, trace attacker actions, and corroborate other evidence. Log analysis often reveals the complete story of an incident.



Data Recovery

Even "deleted" data often remains recoverable. Forensic tools can recover data from formatted drives, damaged storage media, encrypted volumes, and even partially overwritten sectors. This capability is crucial for uncovering evidence that perpetrators attempted to destroy.



Mobile Forensics Overview

Mobile devices contain rich evidence—call logs, messages, location data, app usage, and more. Mobile forensics extracts data from locked devices, analyses app data, recovers deleted information, and examines cloud-synced data. As mobile usage grows, this field becomes increasingly critical.



Cyber Laws & Ethical Responsibility

Importance of Cyber Laws

Cyber laws establish legal frameworks for digital activities, defining what constitutes cyber crime and prescribing punishments. They protect individuals' privacy rights, intellectual property, and digital assets whilst enabling law enforcement to prosecute digital criminals.

In India, the **Information Technology Act, 2000** (amended 2008) governs cyber activities. It addresses cyber crimes like hacking, identity theft, cyber terrorism, and data breaches. Understanding these laws is crucial for any IT professional.

Legal Boundaries

Key principles to remember:

- Unauthorised access is illegal, regardless of intent
- Permission must be explicit and documented
- Data privacy laws protect personal information
- Penalties for cyber crimes are severe—including imprisonment

Ethical Responsibility

As technology professionals, you have ethical obligations beyond legal requirements:

1. **Do no harm:** Never use your skills maliciously
2. **Respect privacy:** Protect user data as you'd want yours protected
3. **Be honest:** Report vulnerabilities responsibly
4. **Stay informed:** Understand legal and ethical implications
5. **Think long-term:** Consider societal impact of your work

Cyber Crime Reporting

In India, cyber crimes can be reported to:

- National Cyber Crime Reporting Portal: cybercrime.gov.in
- Local cyber crime cells
- CERT-In for security incidents





Career Paths in Cyber Security

Cyber security offers diverse, lucrative career opportunities with strong growth prospects. The global shortage of security professionals means high demand and excellent salaries for skilled practitioners.



Security Analyst

Monitor networks for security breaches, investigate violations, and implement security measures. Entry-level role requiring foundational knowledge of networks, operating systems, and security tools.



Ethical Hacker / Penetration Tester

Simulate attacks to identify vulnerabilities before criminals do. Requires deep technical knowledge, creativity, and certification (CEH, OSCP). High-demand role with excellent compensation.



SOC Analyst

Work in Security Operations Centres monitoring, detecting, and responding to security incidents in real-time. Critical 24/7 role protecting organisational assets.



Forensic Investigator

Analyse digital evidence, investigate cyber crimes, and provide expert testimony. Requires technical skills, attention to detail, and understanding of legal procedures.

International Certifications & Skills Roadmap

- CompTIA Security+, Certified Ethical Hacker (CEH)
- CISSP, CISM, Offensive Security Certified Professional (OSCP)
- GIAC certifications, CREST certifications

Essential Skills: Networking, operating systems (Linux/Windows), programming (Python, scripting), security tools, continuous learning mindset



Our Course Value & Final Takeaways

What the Add-On Course Offers

- Hands-on practical labs and real-world scenarios
- Industry-recognised certification preparation
- Expert instruction from practitioners
- Portfolio-building projects
- Direct pathway to security careers

Industry Relevance

- 3.5 million unfilled cyber security jobs globally
- Average security professional salary: ₹6-15 LPA (entry-level)
- Every industry needs security professionals
- Remote work opportunities abundant
- Continuous learning keeps work engaging

Learn Ethically, Think Securely

Remember:

- Security is not just technology—it's a mindset
- Ethical behaviour distinguishes professionals from criminals
- Your skills can protect millions or harm many—choose wisely
- Continuous learning is essential in this ever-evolving field
- The world needs more skilled, ethical security professionals

Questions?

Thank you for your attention. Let's discuss how you can begin your journey in cyber security.